

The sum of the interior angles of a triangle

180° for every triangle, proved with one line drawn parallel to a side — and the exterior angle that follows.

LESSON 86–87

2 × 40 MIN

MATEMATIKA 7, §40

S8 5.2

LESSON CLOCK

10'

25'

20'

25'

5'

The experiment	10 min
The proof	25 min
Finding a missing angle	20 min
The exterior angle	25 min
Homework	5 min

LEARNING OBJECTIVES

- Prove that the angles of a triangle add to 180° .
- Find a missing angle of a triangle.
- State and use the exterior-angle theorem.
- Deduce that a triangle has at most one right or obtuse angle.

TEXTBOOK

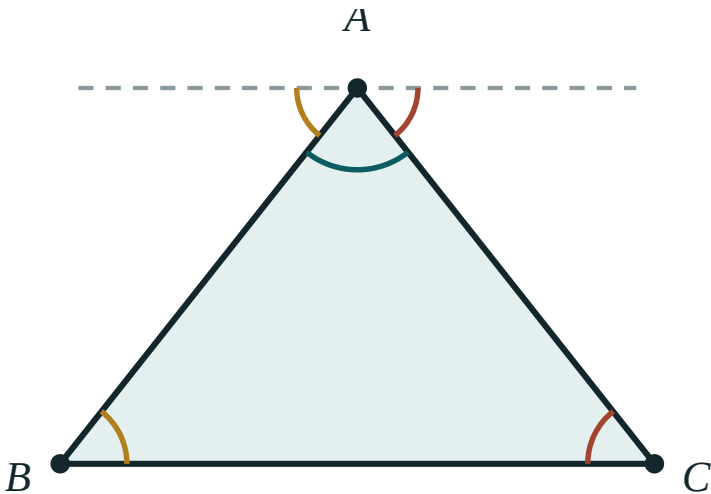
National	pp. 246–253
Cambridge	Stage 8, pp. 48–51
Workbook	Exercise 5.2

01

Explanation

The experiment

Tear the three corners off a paper triangle and lay them side by side: they form a straight angle. Every triangle in the class gives the same result, whatever its shape.



the three angles fill 180°

The three angles laid along a straight line

TRIANGLE	ANGLES	SUM
equilateral	$60^\circ, 60^\circ, 60^\circ$	180°
right	$90^\circ, 55^\circ, 35^\circ$	180°
obtuse	$120^\circ, 40^\circ, 20^\circ$	180°

THE EXPERIMENT IS NOT THE PROOF

Tearing paper checks three triangles; the theorem claims it for all of them. That gap is exactly what a proof closes, and here the proof needs the parallel lines of the last two lessons.

The proof

Through the vertex A draw the line parallel to BC . Now AB and AC are transversals of a pair of parallel lines.

STATEMENT	REASON
draw $\ell \parallel BC$ through A	the axiom of parallels
$\angle 1 = \angle B$	alternate angles, AB a transversal
$\angle 2 = \angle C$	alternate angles, AC a transversal
$\angle 1 + \angle A + \angle 2 = 180^\circ$	angles on a straight line at A
$\angle A + \angle B + \angle C = 180^\circ$	substituting the first two lines

$\angle A + \angle B + \angle C = 180^\circ$

WHERE THE AXIOM HIDES

The proof used the parallel through A — and needed it to be the only one. This theorem is therefore a consequence of the axiom of parallels, not an independent fact.

Finding a missing angle

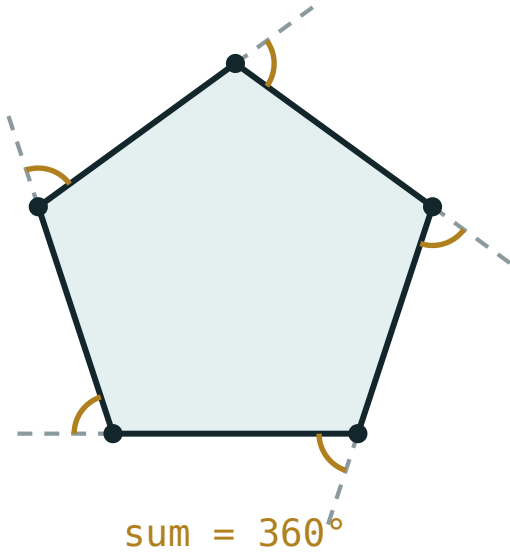
GIVEN	WORKING	THIRD ANGLE
$50^{\circ}, 60^{\circ}$	$180 - 110$	70°
$90^{\circ}, 35^{\circ}$	$180 - 125$	55°
isosceles, apex 40°	$(180 - 40) \div 2$	70° each
angles in ratio $2 : 3 : 4$	$9x = 180$	$40^{\circ}, 60^{\circ}, 80^{\circ}$

THREE CONSEQUENCES, FREE OF CHARGE

A triangle has at most one right angle and at most one obtuse angle, since two of them already reach 180° ; and an equiangular triangle has three angles of 60° .

The exterior angle

Produce one side. The angle outside the triangle beside $\angle C$ is an **exterior angle**, and it equals the sum of the two interior angles not next to it.



An exterior angle equals the two remote interior angles

$\angle_{ext} = \angle A + \angle B$

STATEMENT	REASON
$\angle_{ext} + \angle C = 180^\circ$	angles on a straight line
$\angle A + \angle B + \angle C = 180^\circ$	the angle sum
$\angle_{ext} = \angle A + \angle B$	comparing the two lines

THE TWO REMOTE ANGLES, NOT ALL THREE

An exterior angle equals the sum of the two interior angles at the other vertices. Adding $\angle C$ as well is the standard slip, and it always gives an answer over 180° .

02

Worked examples

EXAMPLE 1 Two angles of a triangle are 50° and 60° . Find the third.

1

$50^\circ + 60^\circ = 110^\circ$

2

$180^\circ - 110^\circ = 70^\circ$

ANSWER

70°

EXAMPLE 2 The angles of a triangle are in the ratio 2 : 3 : 4. Find them.

- ① Let them be $2x$, $3x$, $4x$.
- ② $2x + 3x + 4x = 180^\circ$
- ③ $9x = 180^\circ$, so $x = 20^\circ$.
- ④ The angles are 40° , 60° , 80° .

ANSWER

40° , 60° , 80°

EXAMPLE 3 An exterior angle of a triangle is 120° and one remote interior angle is 45° . Find the other two interior angles.

- ① The remote pair sums to the exterior angle.
- ② $120^\circ - 45^\circ = 75^\circ$
The second remote angle.
- ③ The adjacent angle is $180^\circ - 120^\circ = 60^\circ$.
- ④ Check: $45^\circ + 75^\circ + 60^\circ = 180^\circ$ ✓

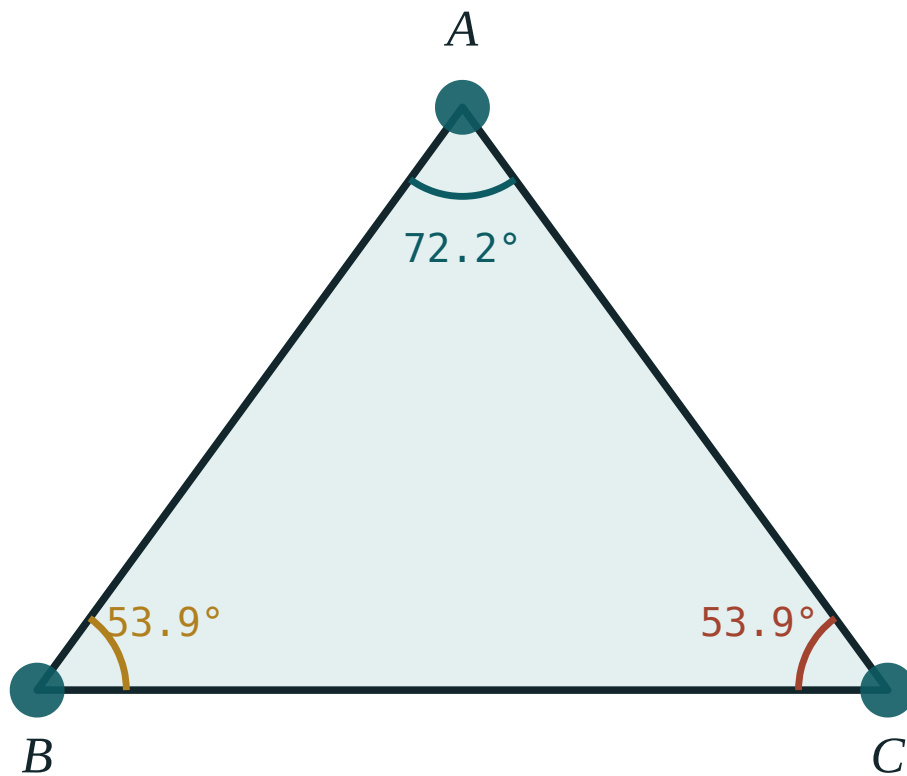
ANSWER

75° and 60°

03 Interactive model

Have the class tear the corners first and only then draw the parallel through the vertex; the proof lands as the explanation of something they have already seen happen.

Drag a vertex — the sum never moves



$\angle A$ 72.2°

$\angle B$ 53.9°

$\angle C$ 53.9°

sum 180°

longest side BC

largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Interior angle	Ichki burchak	Внутренний угол
Exterior angle	Tashqi burchak	Внешний угол
Angle sum	Burchaklar yig'indisi	Сумма углов
Auxiliary line	Yordamchi chiziq	Вспомогательная линия
Theorem	Teorema	Теорема
Acute-angled	O'tkir burchakli	Остроугольный
Obtuse-angled	O'tmas burchakli	Тупоугольный
Equiangular	Teng burchakli	Равноугольный

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The angles of any triangle add to:

The proof draws through a vertex a line:

Two angles are 90° and 35° ; the third is:

A triangle can have at most how many right angles?

1

2

3

any number

An exterior angle equals:

the adjacent angle

the sum of the two remote angles

the sum of all three

180°

An equiangular triangle has angles of:

45°

60°

90°

30°

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 50° and 60° : the third angle

ANSWER 70°

2. 90° and 35°

ANSWER 55°

3. 120° and 40°

ANSWER 20°

4. Each angle of an equiangular triangle

ANSWER 60°

5. Isosceles, apex 40° : the base angles

ANSWER 70° each

6. Isosceles, base angle 50° : the apex

ANSWER 80°

7. Exterior 110° : the adjacent interior angle

ANSWER 70°

Medium · the standard question

7 PROBLEMS

1. Angles in the ratio 2 : 3 : 4

ANSWER $40^\circ, 60^\circ, 80^\circ$

2. Angles in the ratio 1 : 2 : 3

ANSWER $30^\circ, 60^\circ, 90^\circ$

3. Exterior 120° , one remote angle 45° : the other

ANSWER 75°

4. A right triangle with one angle 37° : the third

ANSWER 53°

5. Can a triangle have two obtuse angles?

ANSWER No — they already exceed 180°

6. The angles are $x, 2x, 3x$: find x

ANSWER $x = 30^\circ$

7. Exterior angles of a triangle add to

ANSWER 360°

Hard · stretch and reason

7 PROBLEMS

1. One angle is 20° more than the second and 40° more than the third: find all three

ANSWER $80^\circ, 60^\circ, 40^\circ$

2. An exterior angle is 3 times its adjacent interior angle: find both

ANSWER 135° and 45°

3. An isosceles triangle has an exterior angle of 100° at the apex: find all angles

ANSWER Apex 80° , base angles 50° each

4. The bisectors of $\angle B$ and $\angle C$ meet at I : find $\angle BIC$ if $\angle A = 60^\circ$

ANSWER 120°

5. Prove that a triangle has at most one obtuse angle

ANSWER Two obtuse angles exceed 180° on their own

6. Angles $2x - 10$, $x + 30$, $3x + 10$: find x

ANSWER $x = 25$

7. Why does the proof need the axiom of parallels?

ANSWER It uses the unique parallel through the vertex

07 Homework

Homework — 5 tasks

Check every answer by adding your three angles back to 180° .

- Two angles are 72° and 43° . Find the third.
- The angles are in the ratio $3 : 4 : 5$. Find them.
- Write out the proof of the angle sum in statement-and-reason form.
- An exterior angle is 115° and one remote interior angle is 52° . Find the other two interior angles.
- Explain why a triangle cannot have two right angles.